# Laboration #4 for higher grade

Let the RPI send to a PC with Matlab/Simulink their waypoints, estimated positions and uncertainty. The waypoints estimated positions and uncertainty should be visualized with a plot in runtime. The position and uncertainty should also be saved to file for later evaluation.

- Program the RPI so it sends waypoints estimated positions and uncertainty with at least 10 times/sec by using "socket programming" with either TCP or UDP to a PC.
- Matlab is just used for visualization and storing data.

The examinations will be done in these steps:

1) Supervisor will give an endpoint in x, y, and an angle.

2) The endpoint can then be programmed "hard coded" or be given as an argument to the RPI.

3) The RPI should then create a number of waypoints in cartesian format with a certain sample time from a starting point to the end. (with speed profile)

4) The robot starts to move and send data to Matlab.

5) Matlab plots the waypoints, estimated positions and the uncertainty while moving. It should show clearly within the plot that the Kalman-filter is used, and it works well.

6) Send in the RPI/Matlab code, data and a screen plot.

Good Luck!